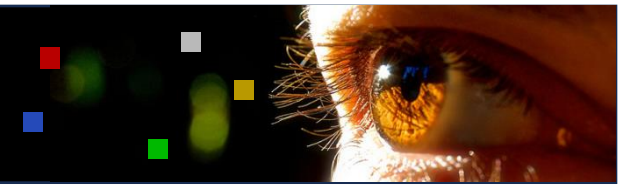
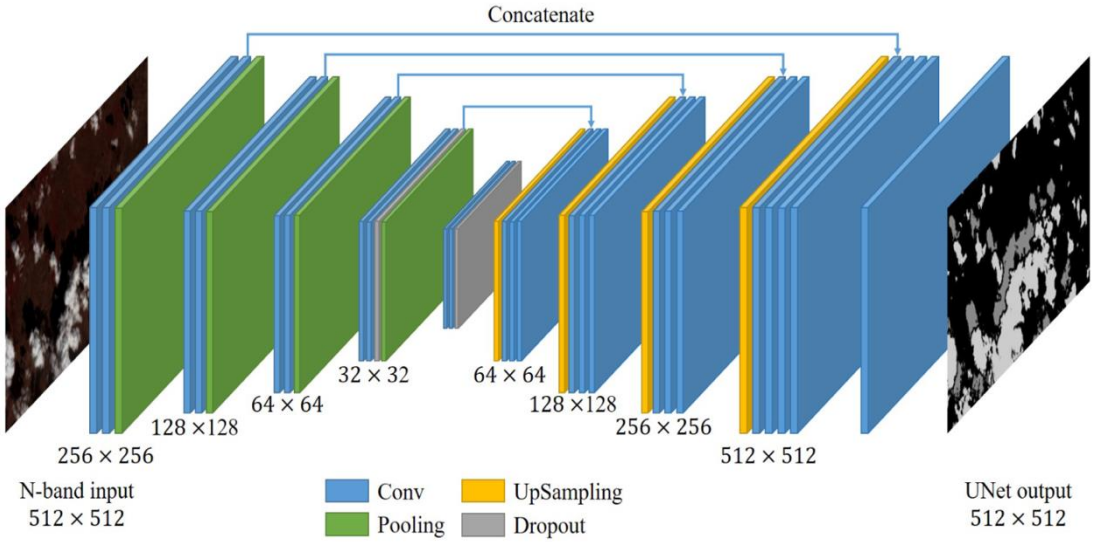
Visión por Computador

Versión 2026



UNet

(CAPÍTULO 3 – CV03E)



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U-Net: Convolutional Networks for Biomedical Image Segmentation

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Abstract. There is large consent that successful training of deep networks requires many thousand annotated training samples. In this paper, we present a network and training strategy that relies on the strong use of data augmentation to use the available annotated samples more efficiently. The architecture consists of a contracting path to capture context and a symmetric expanding path that enables precise localization. We show that such a network can be trained end-to-end from very few images and outperforms the prior best method (a sliding-window convolutional network) on the ISBI challenge for segmentation of neuronal structures in electron microscopic stacks. Using the same network trained on transmitted light microscopy images (phase contrast and DIC) we won the ISBI cell tracking challenge 2015 in these categories by a large margin. Moreover, the network is fast. Segmentation of a 512x512 image takes less than a second on a recent GPU. The full implementation (based on Caffe) and the trained networks are available at http://lmb.informatik.uni-freiburg.de/people/ronneber/u-net.



Olaf Ronneberger

U-net: Convolutional networks for biomedical image segmentation

Authors	Olaf Ronneberger, Philipp Fischer, Thomas Brox
Publication date	2015/10/5
Book	International Conference on Medical image computing and computer-assisted intervention
Pages	234-241
Publisher	Springer international publishing
Description	There is large consent that successful training of deep networks requires many thousand annotated training samples. In this paper, we present a network and training strategy that relies on the strong use of data augmentation to use the available annotated samples more efficiently. The architecture consists of a contracting path to capture context and a symmetric expanding path that enables precise localization. We show that such a network can be trained end-to-end from very few images and outperforms the prior best method (a sliding-window convolutional network) on the ISBI challenge for segmentation of neuronal structures in electron microscopic stacks. Using the same network trained on transmitted light microscopy images (phase contrast and DIC) we won the ISBI cell tracking challenge 2015 in these categories by a large margin. Moreover, the network is fast. Segmentation of a 512x512 image takes less …
Total citations	Cited by 153236

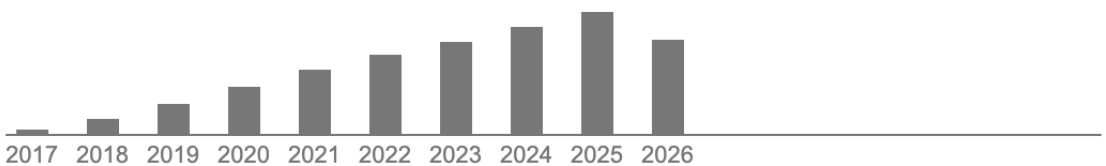


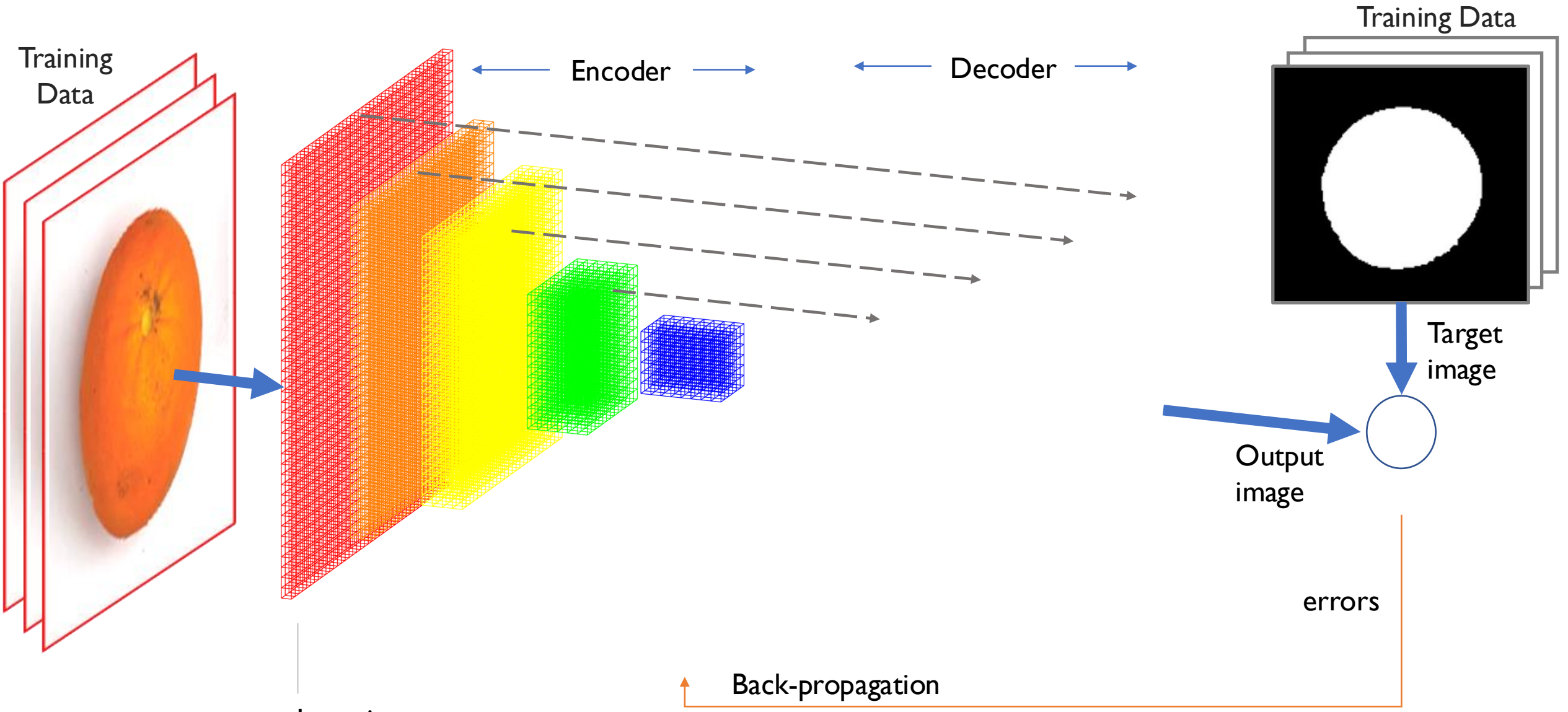
Image Processing – Problem

> Get a target output image from an input image



Image Processing – Solution

Deep Learning Model: U-Net



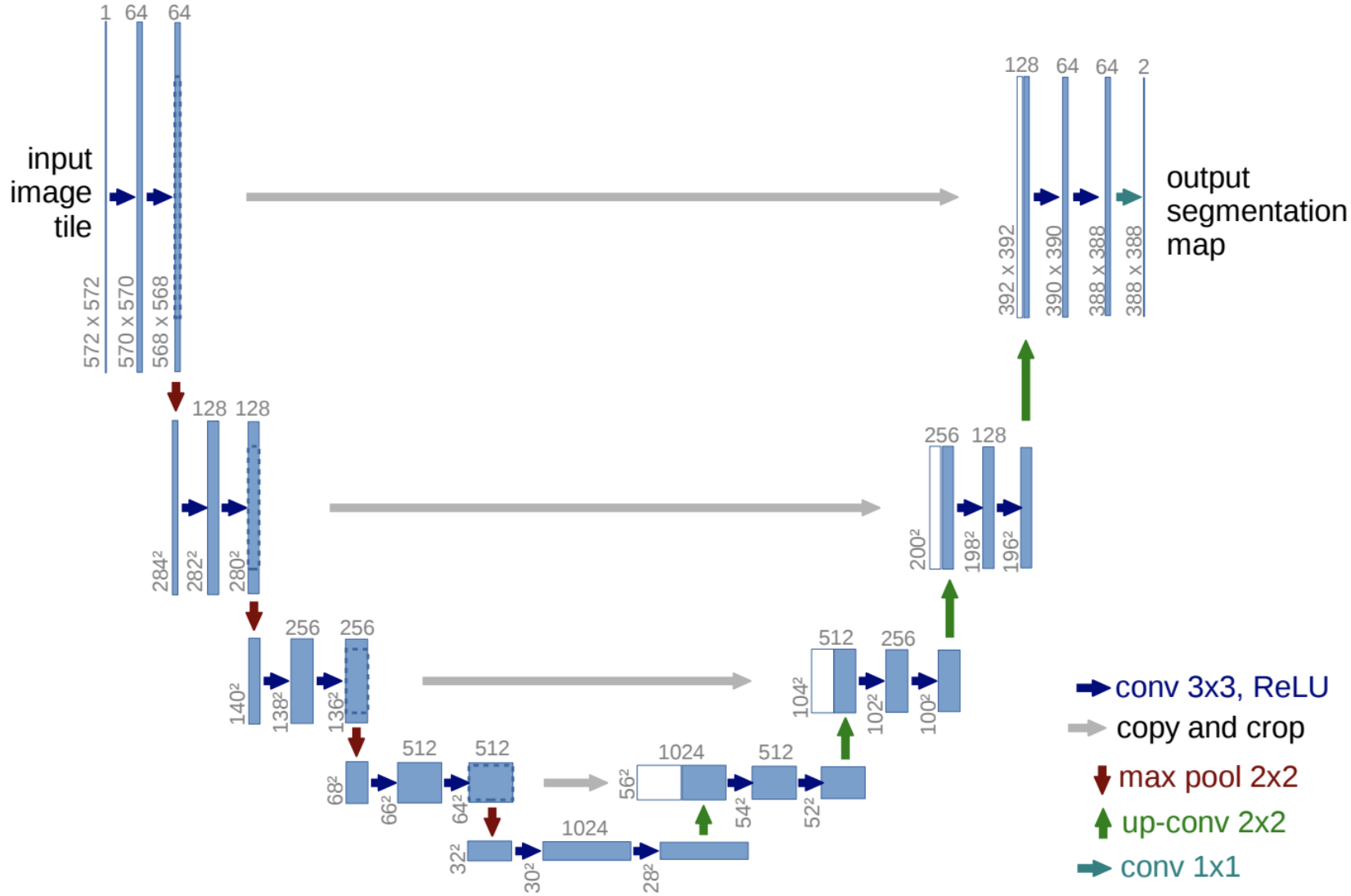
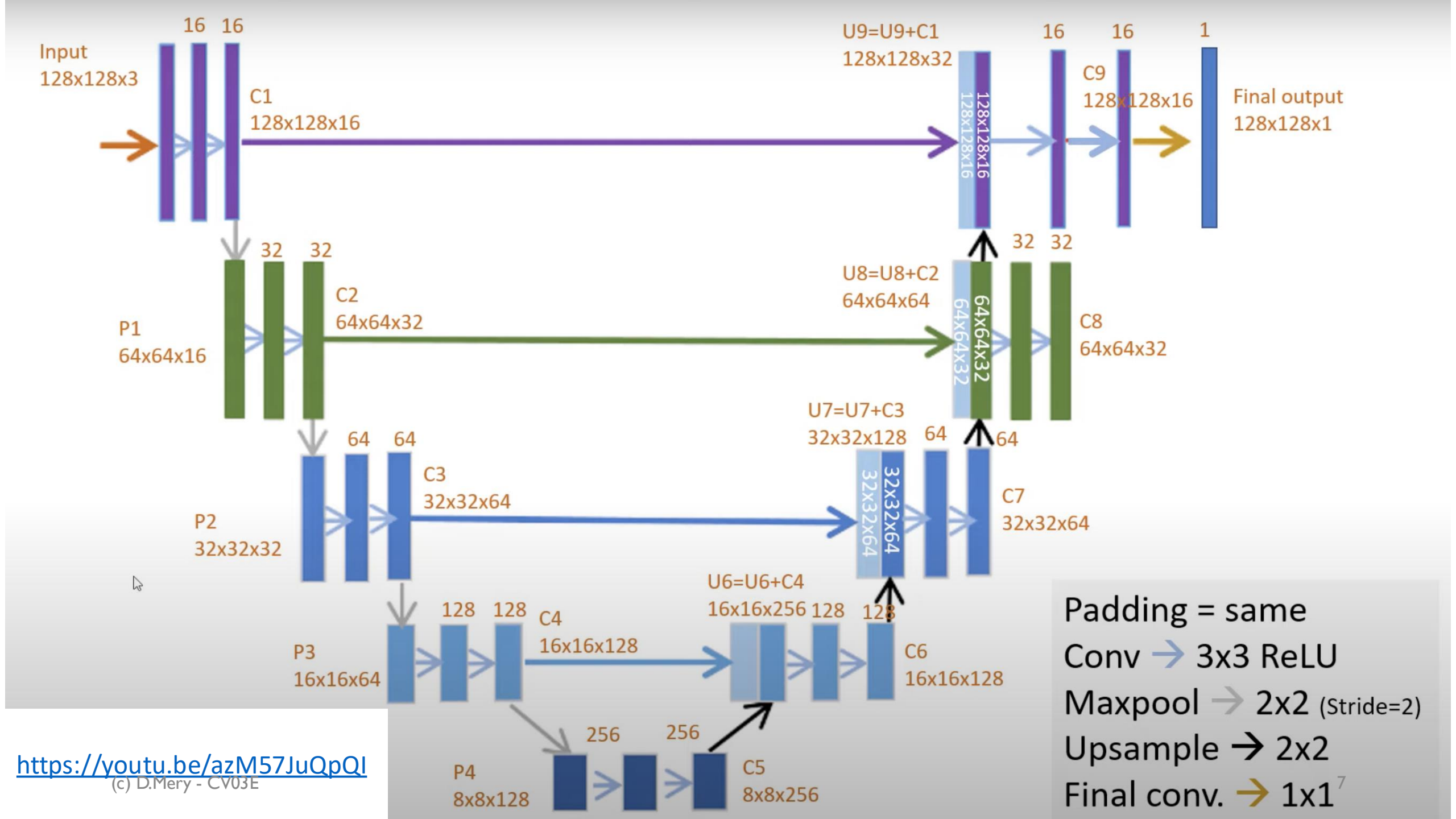


Fig. 1. U-net architecture (example for 32x32 pixels in the lowest resolution). Each blue box corresponds to a multi-channel feature map. The number of channels is denoted on top of the box. The x-y-size is provided at the lower left edge of the box. White boxes represent copied feature maps. The arrows denote the different operations.



https://youtu.be/azM57JuQpQI

Image Segmentation Example

Güiñas Segmentation



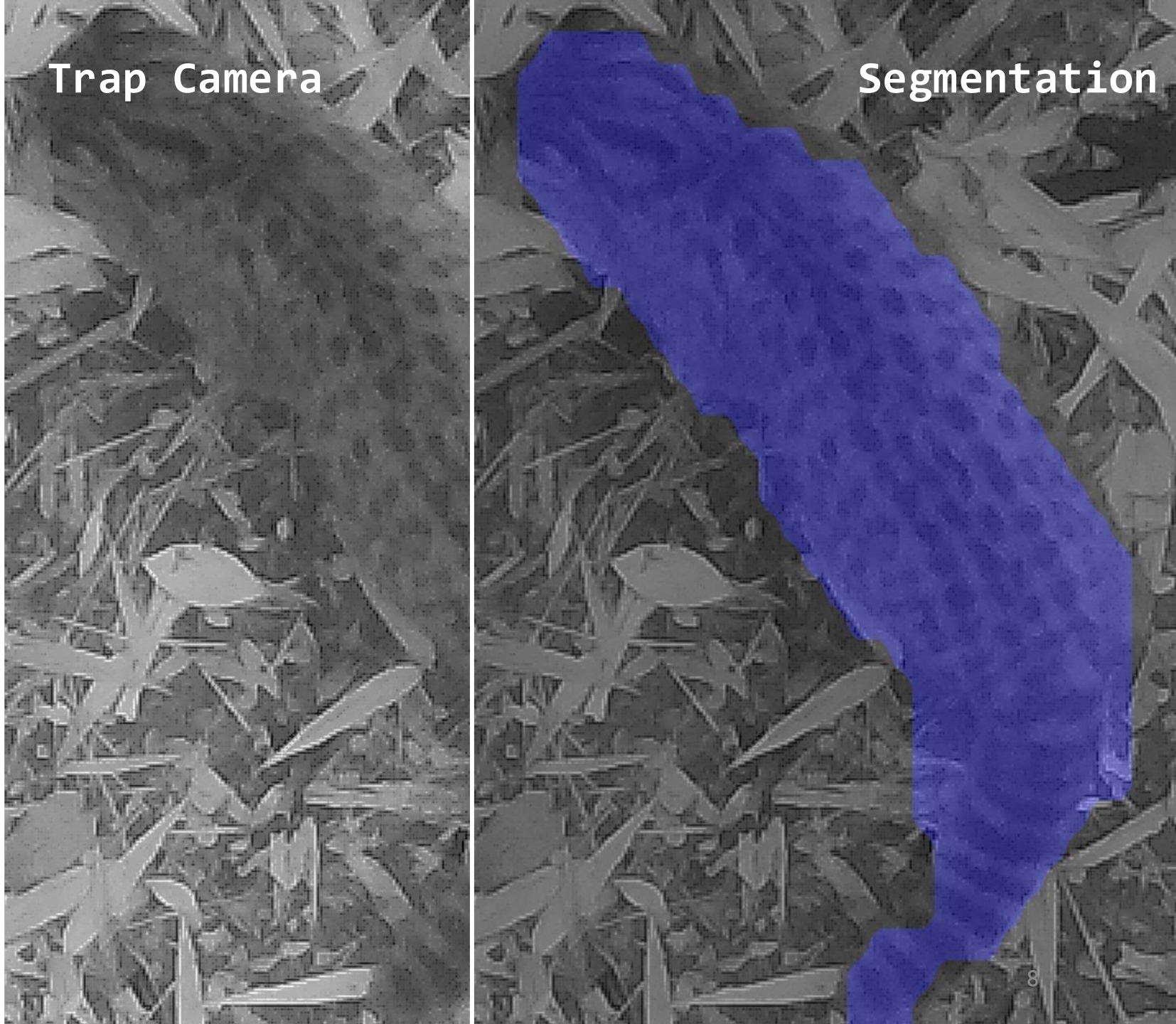


Image Segmentation Example

Skin Lesion Segmentation

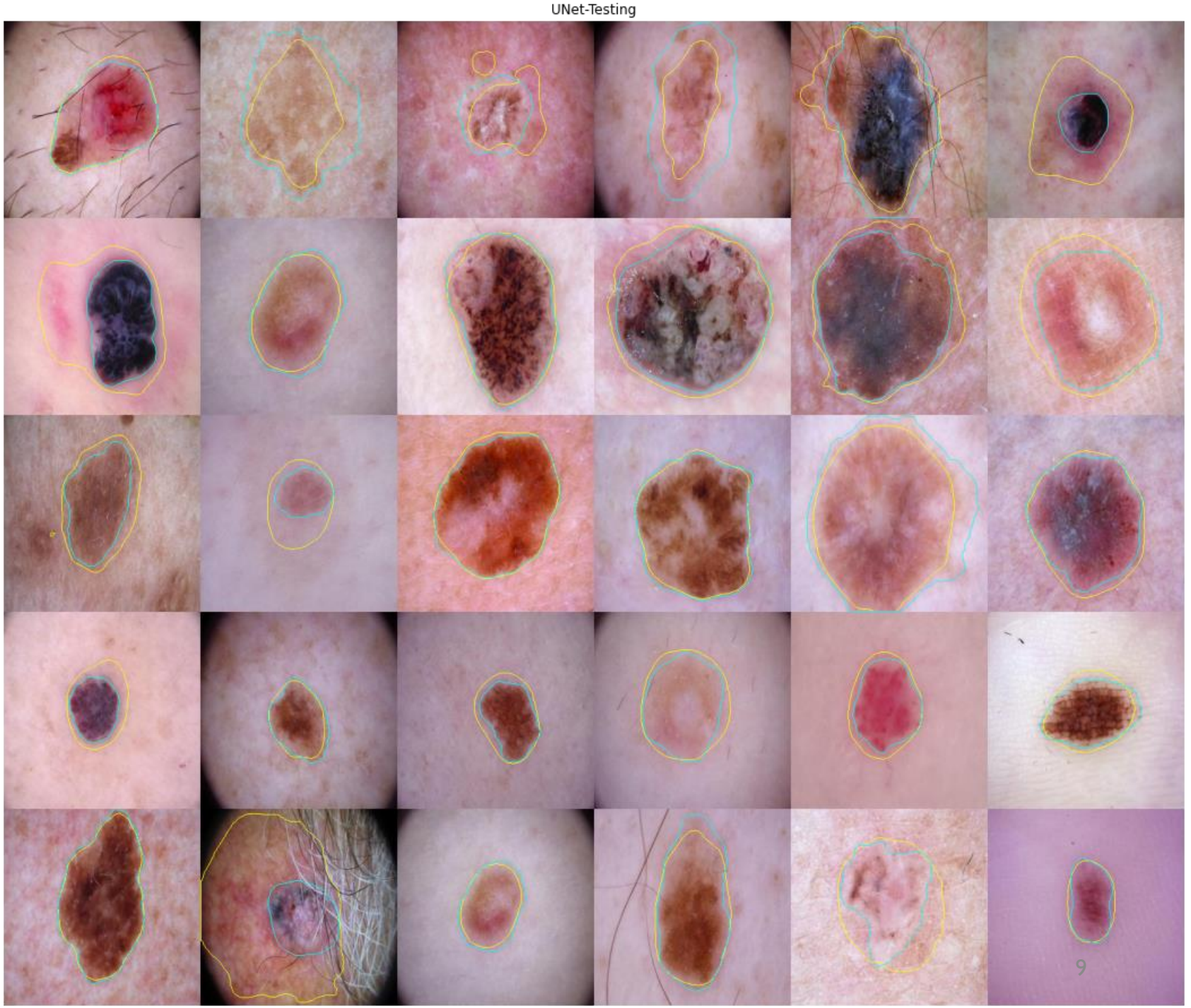


Image Segmentation Example

Breast Tumor Segmentation

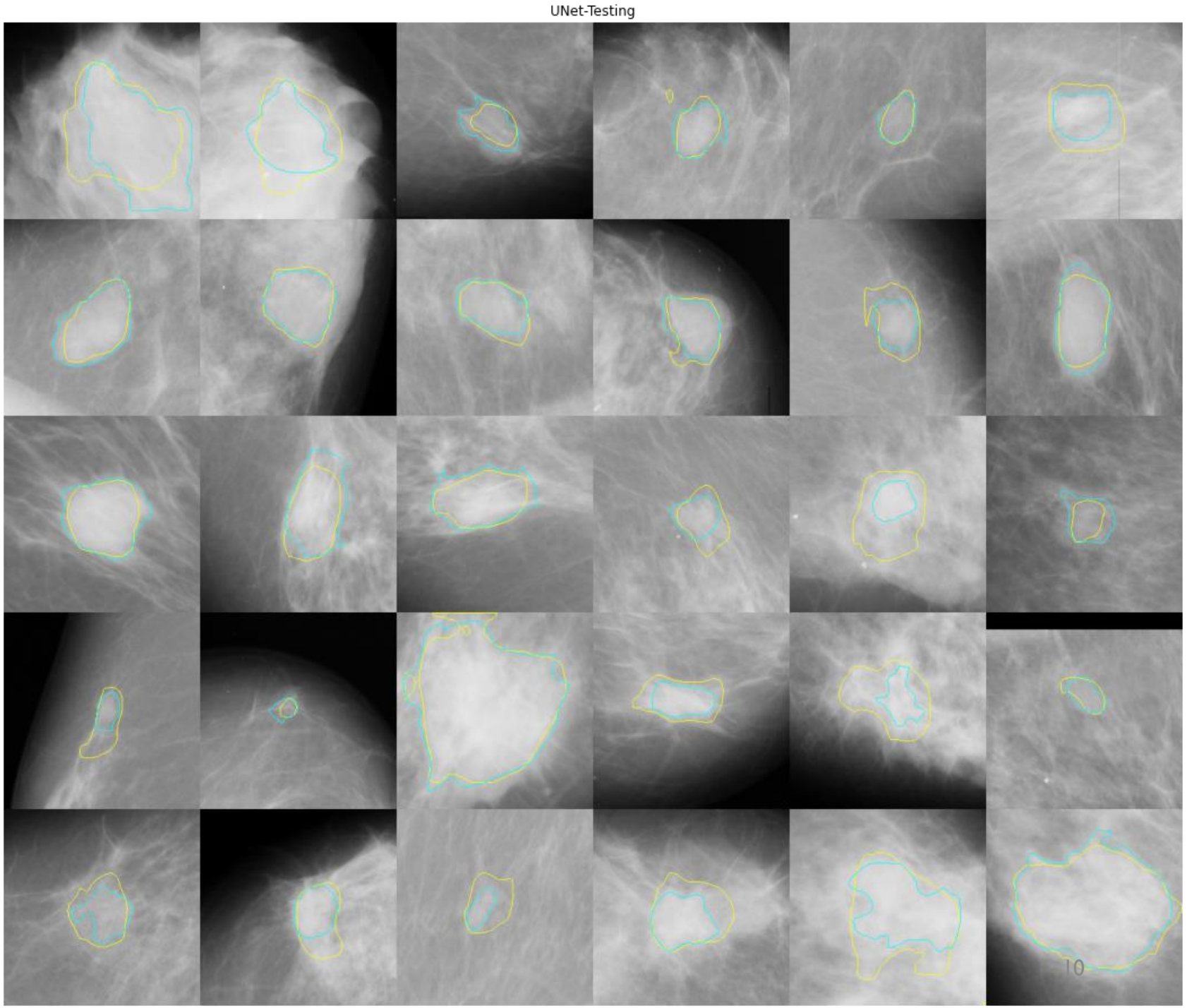
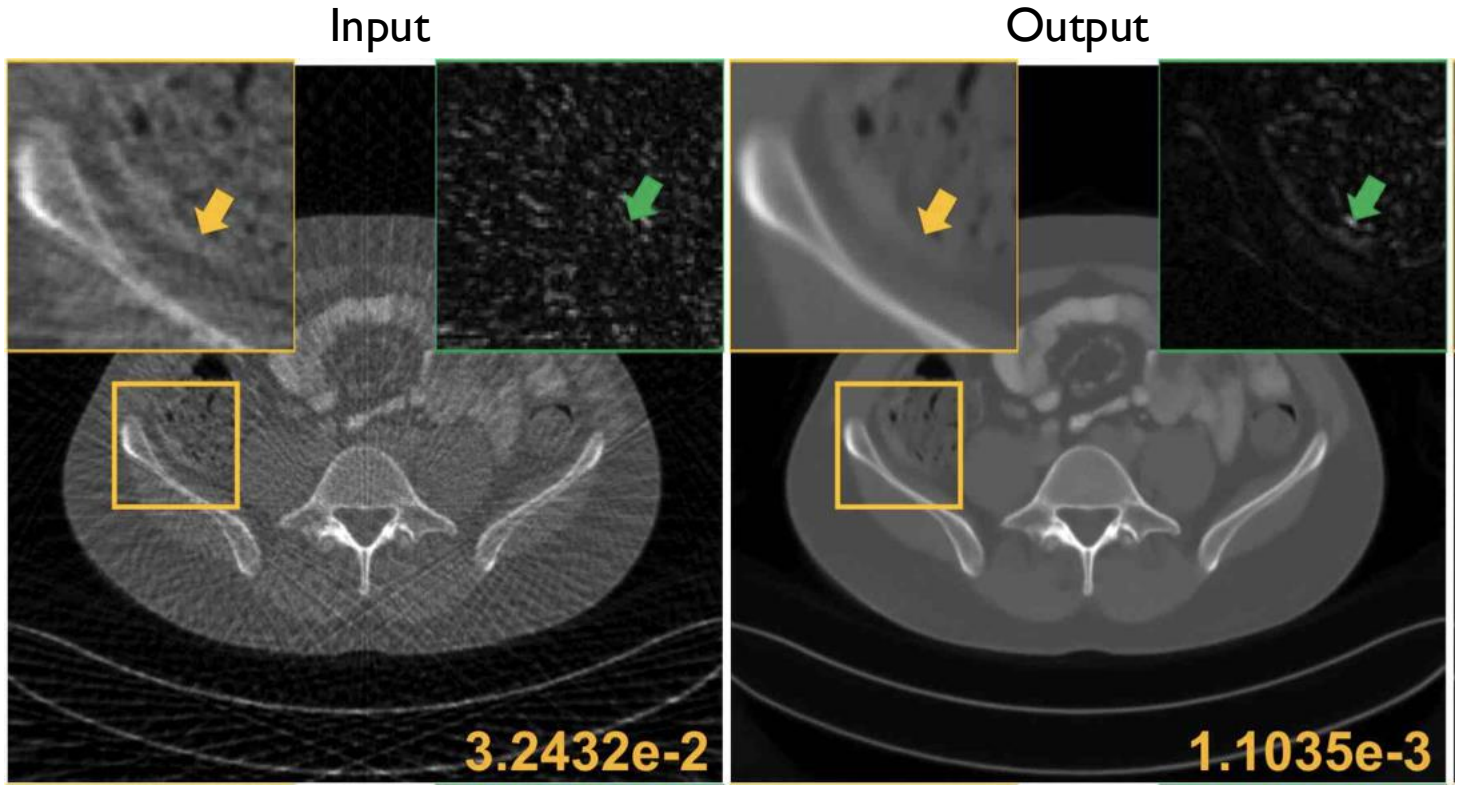


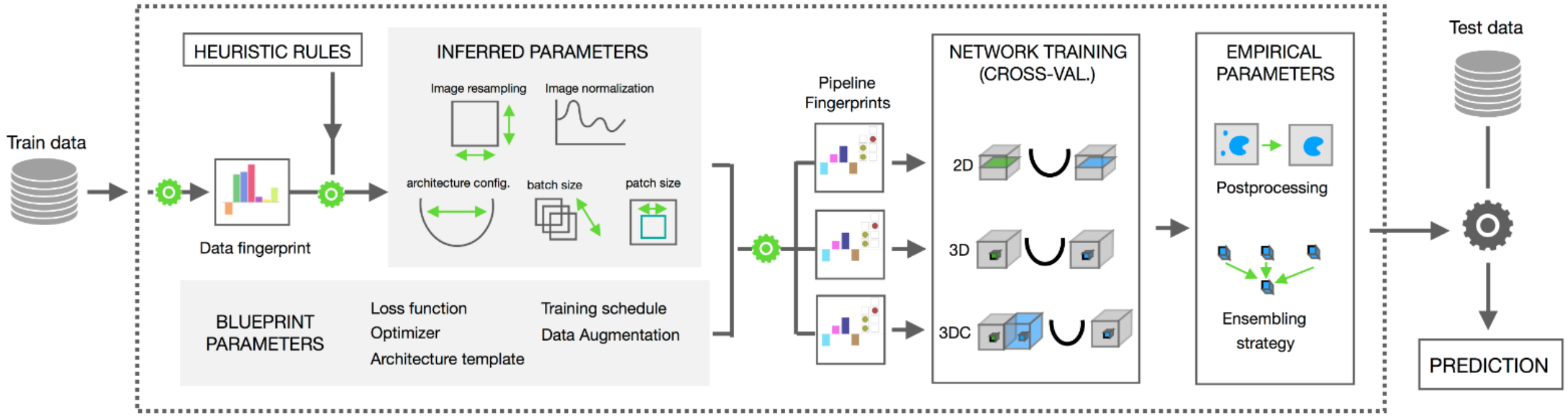
Image Denoising
Example

MRI reconstruction



Han, Yoseob, and Jong Chul Ye. "Framing U-Net via deep convolutional framelets: Application to sparse-view CT." *IEEE transactions on medical imaging* 37.6 (2018): 1418-1429.

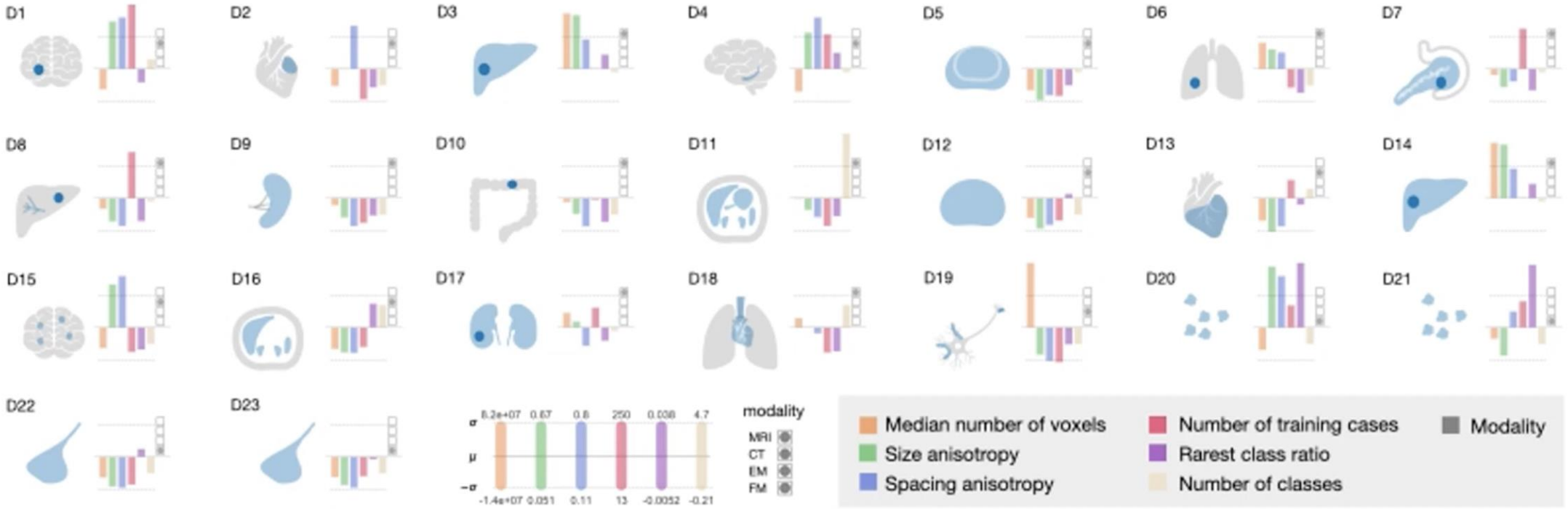
Automated Design of Unet: nnUNet



References:

https://youtu.be/fc2nGJC3mUY
https://medium.com/miccai-educational-initiative/nnu-net-the-no-new-unet-for-automatic-segmentation-8d655f3f6d2a
https://github.com/MIC-DKFZ/nnUNet

Dataset Diversity in Biomedical Images



D1	MSD - Brain Tumor (edema, necrosis, enhancing tumor)	D9	MSD - Spleen (spleen)	D17	KiTS (kidneys, kidney tumor)
D2	MSD - Heart (left atrium)	D10	MSD - Colon (colon cancer)	D18	SegTHOR (heart, aorta, esophagus, trachea)
D3	MSD - Liver (liver, liver tumor)	D11	BCV-Abdomen (13 abdominal organs)	D19	CREMI (synaptic cleft)
D4	MSD - Hippocampus (anterior h., posterior h.)	D12	Promise12 (prostate)	D20	CTC - Fluo-N2DH-SIM+ (HL60 nuclei)
D5	MSD - Prostate (peripheral zone, transition zone)	D13	ACDC (left ventricle, right ventr., myocard.)	D21	CTC - Fluo-N3DH-SIM+ (HL60 nuclei)
D6	MSD - Lung (lung nodules)	D14	LiTS (liver, liver tumor)	D22	CTC - Fluo-C3DH-A549 (A549 cell)
D7	MSD - Pancreas (pancreas, pancreas tumor)	D15	MSLes (ms lesions)	D23	CTC - Fluo-C3DH-A549-SIM (A549 cell)
D8	MSD - Hepatic Vessel (hepatic vessels, liver tumors)	D16	CHAOS (liver, spleen, l/r kidney)		

MSD = Medical Segmentation Decathlon; CTC = Cell Tracking Challenge

Quantitative Results on 19 international Biomedical Challenges

